BoGE Tutorials

Execise correction 2.8

From the tutorial session on 22/04/2026

Fubini and Volume

Exercise 2.26: Area under the curve.

Question 1. Complete and prove the following relation: for all $\Omega \in \mathcal{B}(\mathbb{R})$ and $f \in \mathscr{L}^0(\Omega; \mathbb{R}_+)$,

$$\int_\Omega f(x)\ \mathrm{d}x = \lambda^{\otimes 2}(\{(x,y) \in \mathbb{R}^2 : x \in \Omega \text{ and } \dots\})$$

HINT - *For all* $B \in \mathscr{B}(\mathbb{R}^2),\ \lambda^{\otimes 2}(B) = \int_\Omega \mathbb{1}_B(x,y)\ \lambda^{\otimes 2}(\mathrm{d}x, \mathrm{d}y)$.

A big hint is in the name of the exercise: *Area under the curve.* It is always helpful to do a little sketch of the situation, as in the figure below (Source: Wikimedia Commons). It is often said that the integral measures the "area under the curve", and this exercise serves to justify this phrase. For the first part of the question, we want to link the value of the integral to the area (measure for $\lambda^{\otimes 2}$) under the curve, so we need to complete the condition "$x \in \Omega$ and ..." such that (x, y) is a point under the curve.

Given the above diagram, we can see that for any $(x, y) \in \mathbb{R}^2$, the point (x, y) is in the shaded region if and only if:

$$x \in \Omega \text{ and } 0 \le y \le f(x)$$

We are now going to prove this statement. We denote $D := \{(x, y) \in \mathbb{R}^2 : x \in \Omega \text{ and } 0 \le y \le f(x)\}$, which is in $\mathscr{B}(\mathbb{R}^2)$ (as f is measurable). The given hint lets us write:

$$\lambda^{\otimes 2}(D) = \int_{\mathbb{R}^2} \mathbb{1}_D(x,y)\ \lambda^{\otimes 2}(\mathrm{d}x, \mathrm{d}y)$$

But $\mathbb{1}_D(x,y) = 1$ if and only if we have $x \in \Omega$ and $y \in [0, f(x)]$, so we can write $\mathbb{1}_D(x,y) = \mathbb{1}_\Omega(x) \times \mathbb{1}_{[0,f(x)]}(y)$ for all $(x,y) \in \mathbb{R}^2$. Consequently,

$$\begin{aligned}
\lambda^{\otimes 2}(D) &= \int_{\mathbb{R}^2} \mathbb{1}_\Omega(x)\mathbb{1}_{[0,f(x)]}(y)\ \lambda^{\otimes 2}(\mathrm{d}x, \mathrm{d}y) \\
&= \int_{\mathbb{R}} \int_{\mathbb{R}} \mathbb{1}_\Omega(x)\mathbb{1}_{[0,f(x)]}(y)\ \mathrm{d}y\ \mathrm{d}x && \text{by Fubini} \\
&= \int_{\mathbb{R}} \mathbb{1}_\Omega(x) \left(\int_{\mathbb{R}} \mathbb{1}_{[0,f(x)]}(y)\ \mathrm{d}y \right)\ \mathrm{d}x && \text{since } \int_\Omega \mathbb{1}_A(x) g(x)\ \mathrm{d}x = \int_A g(x)\ \mathrm{d}x \\
&= \int_\Omega \left(\int_0^{f(x)} 1\ \mathrm{d}y \right)\ \mathrm{d}x \\
&= \int_\Omega f(x)\ \mathrm{d}x
\end{aligned}$$

Question 2. Deduce the area of the half unit disk, *i.e.* $\lambda^{\otimes 2}(D)$ where

$$D = \{(x,y) \in \mathbb{R}^2 : x^2 + y^2 \leq 1 \text{ and } y \geq 0\}$$

HINT - *The change of variables $x = \cos\theta$ may be useful.*

The question starts with *deduce*, meaning that we are going to use the previous question. As we are trying to calculate $\lambda^{\otimes 2}(D)$, the previous question requires us to rewrite D in the form $\{(x,y) \in \mathbb{R}^2 : x \in \Omega \text{ and } 0 \leq y \leq f(x)\}$ for some $\Omega \in \mathscr{B}(\mathbb{R})$ and some $f \in \mathscr{L}^0(\Omega; \mathbb{R}_+)$. A quick sketch of D (which we are already told is the half unit disk) reveals that $\Omega = [-1,1]$ and:

$$f : x \in [-1,1] \mapsto \sqrt{1 - x^2}$$

are suitable (one can manually verify that this does redefine D).

Applying the previous question to the newfound Ω and f, we have that $\lambda^{\otimes 2}(D) = \int_{-1}^1 \sqrt{1-x^2}\ \mathrm{d}x$. This is the point at which the change of variables offered by the question comes into play. Subject to justifying that this change of variables is well-defined, the integrand $\sqrt{1-x^2}$ would become $\sqrt{1-(\cos\theta)^2} = \sqrt{(\sin\theta)^2} = |\sin\theta|$ (don't forget the absolute value!). It is therefore reasonable to expect that this change of variables can prove useful in calculating the integral.

Let's justify the change of variables: the function

$$\cos \quad : \quad \begin{array}{ccc}]0,\pi[& \to &]-1,1[\\ \theta & \mapsto & \cos\theta \end{array}$$

is one-to-one and onto, so $\cos^{-1} = \arccos$ is well-defined, and each of these functions are $\mathscr{C}^1$ over their respective domains, so cos is a $\mathscr{C}^1$-diffeomorphism. Because cos has real variable and real value, the determinant of the jacobian matrix is simply equal to $|\cos'(x)| = |\sin(x)|$. So:

$$\lambda^{\otimes 2}(D) = \int_0^\pi (\sin\theta)^2\ \mathrm{d}\theta$$

The heart of the exercise does not rest in the evaluation of this integral, but rather the methods used to reach this point. We are still going to calculate its value however. By integration by parts, we have:

$$\begin{aligned}
\lambda^{\otimes 2}(D) &= \int_0^\pi (\sin\theta)^2\ \mathrm{d}\theta \\
&= [-\sin\theta\cos\theta]_0^\pi - \int_0^\pi -(\cos\theta)^2\ \mathrm{d}\theta \\
&= \int_0^\pi (\cos\theta)^2\ \mathrm{d}\theta
\end{aligned}$$

We thus have:
$$\int_0^\pi (\sin\theta)^2 \ \mathrm{d}\theta = \frac{1}{2}\left(\int_0^\pi (\sin\theta)^2 + (\cos\theta)^2 \ \mathrm{d}\theta\right) = \frac{\pi}{2}$$

Exercise 2.27: Area of an ellipse.

Let $\alpha, \beta > 0$ and define
$$E_{\alpha,\beta} = \left\{(u,v) \in \mathbb{R}^2 : \frac{u^2}{\alpha^2} + \frac{v^2}{\beta^2} \leq 1\right\}$$

Question 1. Draw $E_{\alpha,\beta}$.

Left to the reader.

Question 2. a) Denote B the unit ball of center 0 in $(\mathbb{R}^2, \|\cdot\|_2)$. Find $M \in \mathrm{Mat}_2(\mathbb{R})$ such that
$$E_{\alpha,\beta} = \{Mz : z \in B\}$$

This question can be solved through trial and error, knowing that $B = \{(x,y) \in \mathbb{R}^2 : \|(x,y)\|_2 \leq 1\}$. Note that we can also rewrite:
$$E_{\alpha,\beta} = \left\{(u,v) \in \mathbb{R}^2 : \left\|\left(\frac{u}{\alpha}, \frac{v}{\beta}\right)\right\|_2 \leq 1\right\}$$
So if (x,y) is in B, we have that $(\alpha x, \beta y)$ is in $E_{\alpha,\beta}$. We can therefore take M to be:
$$M = \begin{bmatrix} \alpha & 0 \\ 0 & \beta \end{bmatrix}$$
We indeed have, for all (x,y) in $\mathbb{R}^2$, that:
$$\begin{aligned}(x,y) \in E_{\alpha,\beta} &\iff \left\|\left(\frac{x}{\alpha}, \frac{y}{\beta}\right)\right\|_2 \leq 1 \\ &\iff \left(\frac{x}{\alpha}, \frac{y}{\beta}\right) \in B \\ &\iff (x,y) = M\begin{bmatrix} \frac{x}{\alpha} \\ \frac{y}{\beta} \end{bmatrix} \in \{Mz : z \in B\}\end{aligned}$$

Question 2. b) Compute $\det M$.

Because M is diagonal, $\det M = \alpha\beta > 0$.

Question 3. Compute $\lambda^{\otimes 2}(E_{\alpha,\beta})$. *Consider known the value of* $\lambda^{\otimes 2}(B)$.

Using properties of the Lebesgue measure, we have that:
$$\lambda^{\otimes 2}(E_{\alpha,\beta}) = \lambda^{\otimes 2}(\{Mz : z \in B\}) = |\det M\ | \times \lambda^{\otimes 2}(B) = \alpha\beta \times \pi$$
In the drawing of $E_{\alpha,\beta}$, ask yourself what α, β represent. Does the formula we've just found for the area of an ellipse seem coherent? If $\alpha = \beta$, what geometric shape do we obtain, and does this formula still work?

Exercise 2.28: Determinant training 3

Compute the determinant of the following matrices:

Question 4. $\begin{bmatrix} -1 & 1 & 2 \\ 2 & -2 & 0 \\ 1 & -2 & 2 \end{bmatrix}$

Using the formula for the determinant of a 3×3 matrix, we get:

$$\begin{aligned} \det M &= (-1) \times (-2) \times 2 + 2 \times (-2) \times 2 + 1 \times 0 \times 1 \\ &\quad - 2 \times (-2) \times 1 - 1 \times 2 \times 2 - (-1) \times (-2) \times 0 \\ &= 4 - 8 + 4 - 4 = \boxed{-4} \end{aligned}$$

Question 5. $\begin{bmatrix} 0 & \dots & 0 & 1 \\ \vdots & & & \vdots \\ 0 & 1 & \dots & 0 \\ 1 & 0 & \dots & 0 \end{bmatrix} \in \mathrm{Mat}_d(\mathbb{R})$

This matrix can be seen as the identity matrix I_d, but "flipped". Note that we could exchange the top row with the bottom row, then the second row with the second-to-last row, etc... This would give us $\lfloor \frac{d}{2} \rfloor$ row swaps, meaning that the determinant of this matrix is:

$$\det M = (-1)^{\lfloor \frac{d}{2} \rfloor}$$

We can verify our work by taking small values of d: for $d = 1$, the matrix has determinant 1, which is correct for the above formula. For $d = 2$, one row swap is necessary, so we have a determinant of -1, which still works.

Exercise 2.29: Area of a 'cone'

Let $h \in \mathbb{R}_+$, $B \in \mathscr{B}(\mathbb{R}^2)$ and denote

$$C = \left\{ (tx, ty, (1-t)h) \mid t \in [0,1], (x,y) \in B \right\}$$

We assume that $C \in \mathscr{B}(\mathbb{R}^3)$.
Question 6. Draw C.

Left to the reader (this is a good visual exercise, so do it!). To simplify a first drawing, assume that B is the unit disk of $\mathbb{R}^2$.

Question 7. Compute $\lambda^{\otimes 3}(C)$ via the following method:

(i) write $\lambda^{\otimes 3}(C) = \int_{\mathbb{R}^3} f(u,v,w)\ \lambda^{\otimes 3}(\mathrm{d}u, \mathrm{d}v, \mathrm{d}w)$ for some function f,

(ii) justify and apply Fubini's theorem

(iii) successively perform the following changes of variables: $w = (1-t)h,\ u = tx$, and $v = ty$.

(i) A good trick to know is that $\lambda^{\otimes d}(S) = \int_{\mathbb{R}^d} \mathbb{1}_S(x)\ \mathrm{d}x$ (this is an immediate consequence of the definition of the Lebesgue integral). Applying this to the current exercise grants:

$$\lambda^{\otimes 3}(C) = \int_{\mathbb{R}^3} \mathbb{1}_C(u,v,w)\ \lambda^{\otimes 3}(\mathrm{d}u, \mathrm{d}v, \mathrm{d}w)$$

where $\mathbb{1}_C$ is measurable because $C \in \mathscr{B}(\mathbb{R}^3)$.

(ii) The function $\mathbb{1}_C$ is positive and measurable over $\mathbb{R}^3$. By Fubini's theorem, we have:

$$\lambda^{\otimes 3}(C) = \int_{\mathbb{R}} \int_{\mathbb{R}} \int_{\mathbb{R}} \mathbb{1}_C(u,v,w)\ \mathrm{d}w\ \mathrm{d}u\ \mathrm{d}v$$

You might be wondering how we know which order to integrate in (why, for example, didn't we write du dv dw instead?). Fubini's theorem allows us to write the variables in any order we want. However, the following step tells us in which order we should perform a change of variables, so it is reasonable to expect that the same order is best adapted to the integration.

(iii) Let's (properly) justify the change of variables $w = (1-t)h$ in the inner integral. We first suppose that $h > 0$: we will treat the general case later. We denote:

$$\begin{array}{rccc} \varphi : &]0,1[& \to &]0,h[\\ & t & \mapsto & (1-t)h \end{array}$$

φ is evidently $\mathscr{C}^1$ over $]0,1[$, and it is a bijection. Moreover, it's reciprocal can be written

$$\begin{array}{rccc} \varphi^{-1} : &]0,h[& \to &]0,1[\\ & z & \mapsto & 1-\frac{|z|}{|h|} \end{array}$$

where $\|\cdot\|_2$ is the euclidean norm over $\mathbb{R}^3$. (You can verify that φ^{-1} is the reciprocal function of φ). This expression of φ^{-1} is not necessary to write, but helps understand why φ^{-1} is $\mathscr{C}^1$. Thus, φ is a $\mathscr{C}^1$-diffeomorphism and the variable change is therefore justified. The jacobian matrix of φ is equal to $-h$, so we have for fixed $u, v \in \mathbb{R}$:

$$\int_{\mathbb{R}} \mathbb{1}_C(u,v,w)\ \mathrm{d}w = \int_0^h \mathbb{1}_C(u,v,w)\ \mathrm{d}w = h \int_0^1 \mathbb{1}_C(u,v,(1-t)h)\ \mathrm{d}t$$

where the first equality comes from $\mathbb{1}_C(u,v,w) = 0$ whenever $w \notin [0,h]$.

Moving on to the second variable change, $u = tx$, we would like t to be fixed for the corresponding integral. So, we are going to apply Fubini again to obtain:

$$\lambda^{\otimes 3}(C) = h \int_0^1 \int_{\mathbb{R}} \int_{\mathbb{R}} \mathbb{1}_C(u,v,(1-t)h)\ \mathrm{d}u\ \mathrm{d}v\ \mathrm{d}t$$

We can similarly justify the variable change, obtaining this time d$u = t$ dx and thus for fixed $t \in]0,1[$ and $v \in \mathbb{R}$:

$$\int_{\mathbb{R}} \mathbb{1}_C(u,v,(1-t)h)\ \mathrm{d}u = t \times \int_{\mathbb{R}} \mathbb{1}_C(tx,v,(1-t)h)\ \mathrm{d}x$$

Similarly, for v:

$$\int_{\mathbb{R}} \int_{\mathbb{R}} \mathbb{1}_C(u,v,(1-t)h)\ \mathrm{d}u\ \mathrm{d}v\ \mathrm{d}t = t^2 \times \int_{\mathbb{R}} \int_{\mathbb{R}} \mathbb{1}_C(tx,ty,(1-t)h)\ \mathrm{d}x\ \mathrm{d}y = t^2 \times \int_{\mathbb{R}} \int_{\mathbb{R}} \mathbb{1}_B(x,y)\ \mathrm{d}x\ \mathrm{d}y$$

Indeed, by the definition of C, the previous variable changes allow us to replace $\mathbb{1}_C(tx,ty,(1-t)h)$ by $\mathbb{1}_B(x,y)$ whenever $t \in [0,1]$. However, by Fubini:

$$\int_{\mathbb{R}} \int_{\mathbb{R}} \mathbb{1}_B(x,y)\ \mathrm{d}x\ \mathrm{d}y = \int_{\mathbb{R}^2} \mathbb{1}_B(z)\ \lambda^{\otimes 2}(\mathrm{d}x, \mathrm{d}y) = \lambda^{\otimes 2}(B)$$

Hence:

$$\lambda^{\otimes 3}(C) = h \times \lambda^{\otimes 2}(B) \times \int_0^1 t^2\ \mathrm{d}t = \boxed{\frac{h}{3}\lambda^{\otimes 2}(B)}$$

Let's now treat the case of $h = 0$. In this case C is contained in a plane of $\mathbb{R}^3$. Thus, the measure of C is 0 for the Lebesgue 3-dimensional measure. For the previously-announced formula above, we obtain 0 on the right-hand side as well, meaning that the formula holds for any $h \in \mathbb{R}_+$.

Let's test our newfound formula $\lambda^{\otimes 3}(C) = \frac{h}{3}\lambda^{\otimes 2}(B)$ when B is a disk of radius r centered at 0. Using the drawing you did for question 1, C is a right circular cone (the "usual" notion of a cone), and the volume of this cone is given by $\lambda^{\otimes 3}(C) = \frac{h}{3}\pi r^2$, which is what we would find with our formula. **END.**